

# **BAAZ & UDAY JKBOSE**

**2026 CLASS 11<sup>TH</sup>**

## **MECHANICAL PROPERTIES OF SOLIDS**

**LECTURE-03** **PHYSICS**

**WAJAHID SIR (Gabbar Sir)**



# Topics To Be Covered

- 1 Revision (dec - 1 and 2)
- 2 Elastic Potential Energy (Bohat easy) # Spring pe
- 3 .. p-Ratim
- 4 ..
- 5 ..



# Recap From 1<sup>st</sup> Lecture

## Properties of Solid

- Maximum IAF 
- minimum IAD 

External Force

Restoring Force

## Elasticity

Steel >>> Rubber

# Property to regain its shape

## Plasticity

# Do not regain its shape

## Stress

- The Restoring force acting per unit Area

$$\text{Stress} = \frac{F_R}{A} = \frac{N}{m^2} \text{ (Pascal)}$$

⇒ Longitudinal Stress / Normal Stress

⇒ Volumetric Stress / Bulk Stress

⇒ Tagential Stress / Shear Stress

$$\text{STRAIN} = \frac{\Delta l}{l}$$

# Recap From 2<sup>nd</sup> Lecture

- Strain =  $\frac{\Delta C}{C_0}$

longitudinal Strain

$$\frac{\Delta l}{l}$$

Hooke's law

$$\text{Stress} = e \text{ Strain}$$

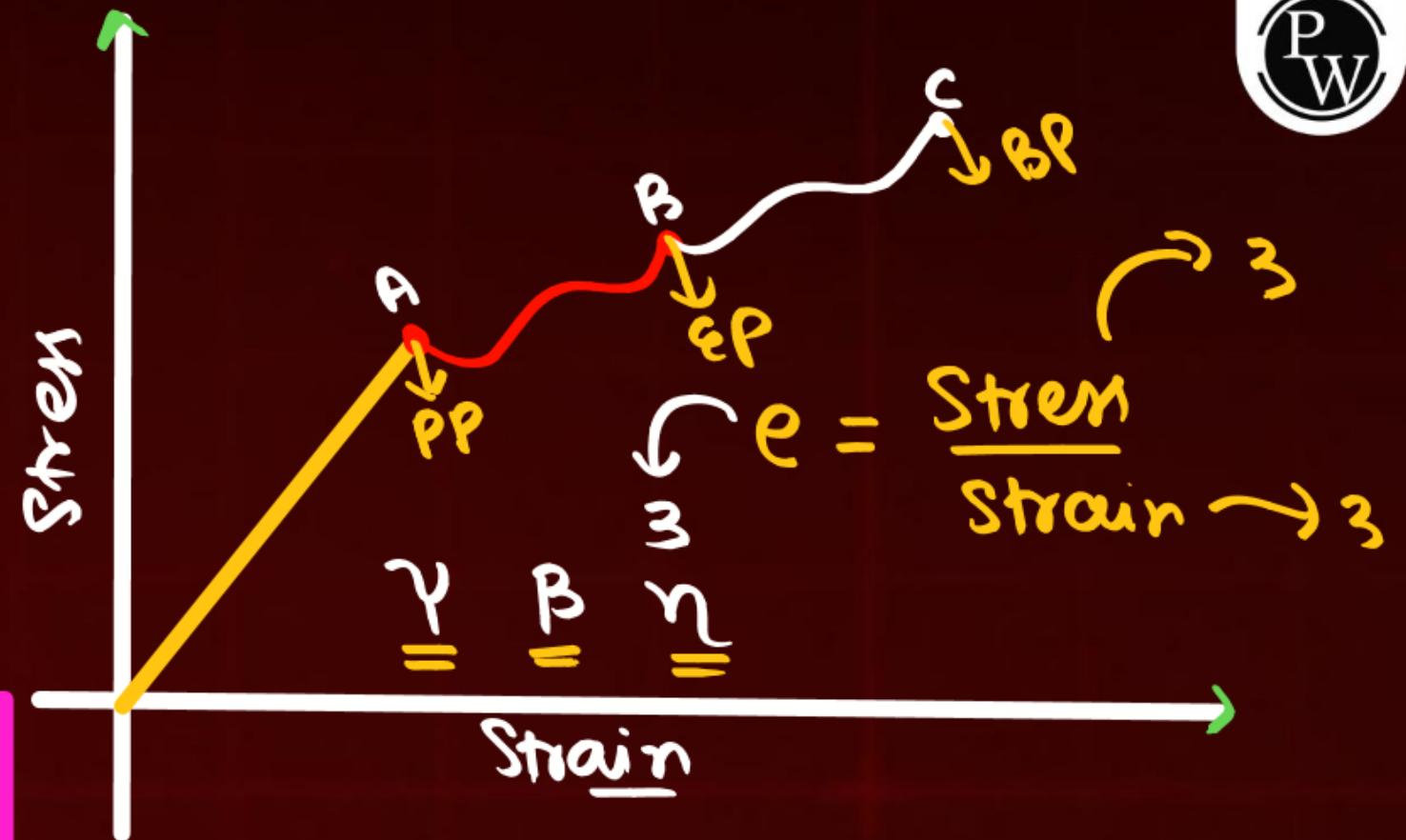
Volumetric Strain

$$\frac{\Delta V}{V}$$

$\rightarrow F_{ext}$

Shear Strain

$$\tan \theta$$



Spring force and Spring constant

$$F_s = -kx$$

Rod force and Rod constant

$$R_R = -k\Delta l$$

$$k = \frac{AY}{l}$$



### lecture no. 3

## Factor effecting the (modulus of elasticity $e$ ) Elasticity

### ① Temperature (2022 Neet)

On increasing the temperature of Elastic material  
its elasticity decreases



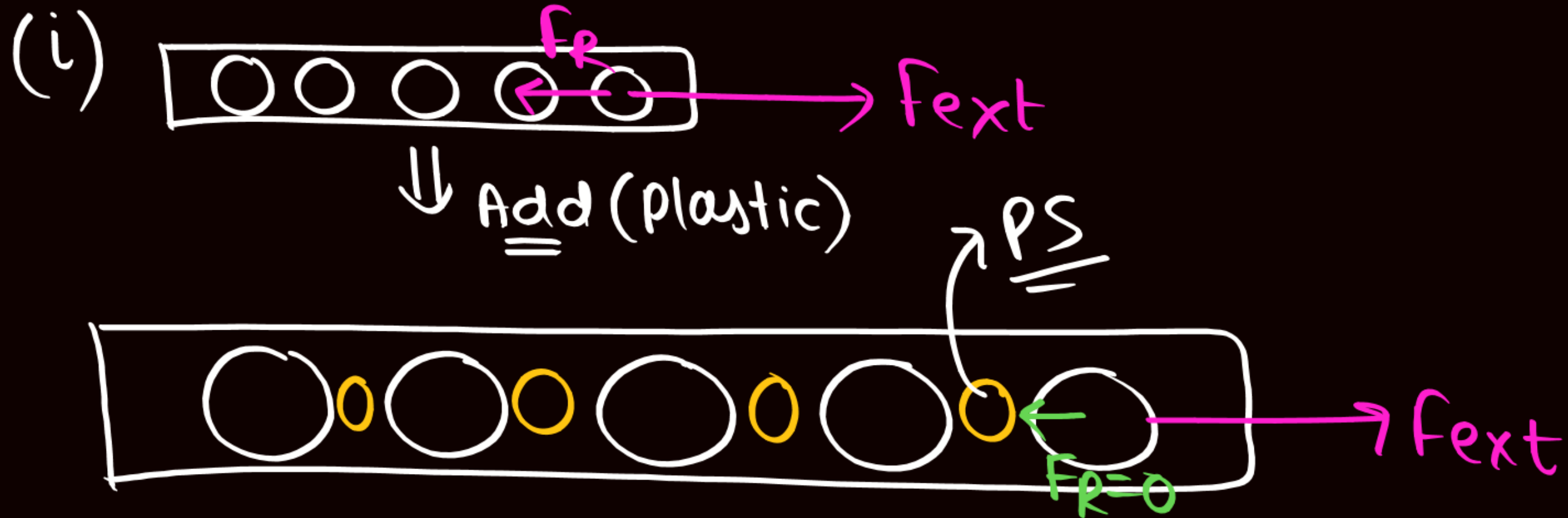
Temp  $\uparrow\uparrow$   $e \downarrow\downarrow$  modulus of  $e \downarrow\downarrow$



$f_R \downarrow\downarrow\downarrow$

## ② Impurities

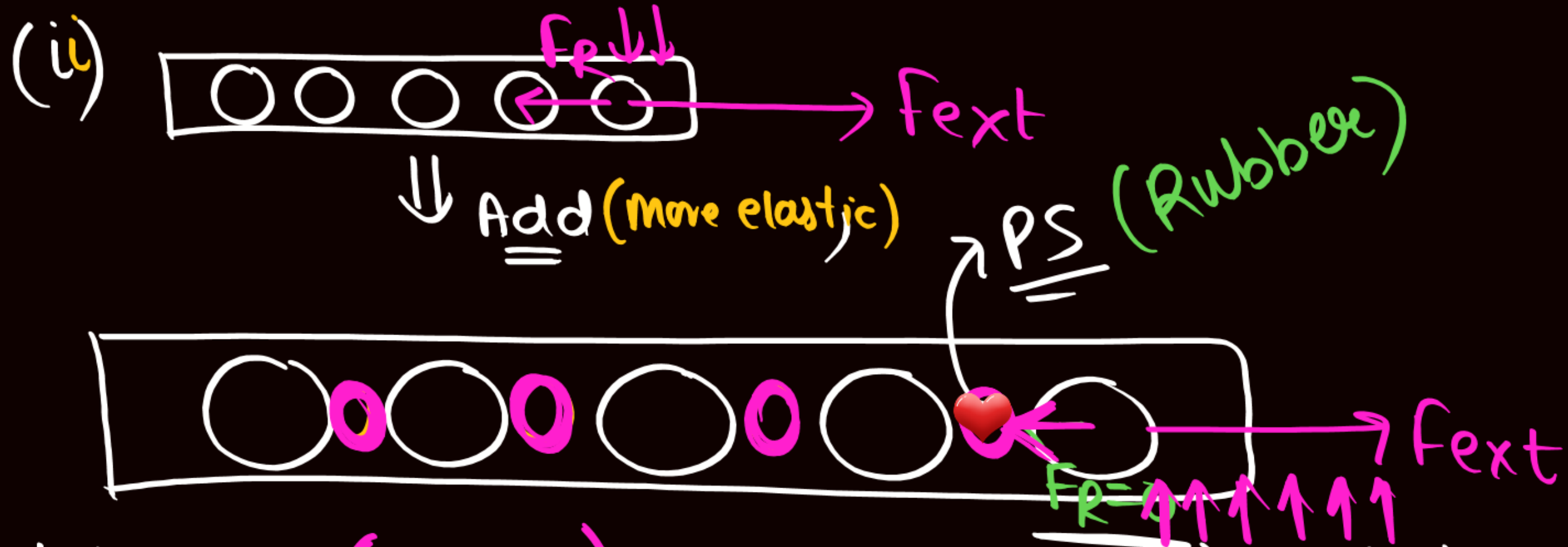
Adding other substance to elastic material



When inelastic (plastic substance) add to the elastic substance its elasticity  $(e)$  decrease.

## ② Impurities.

Adding other substance to elastic material



When (elastic) substance its elasticity increases when (elastic) substance is added to the elastic substance.

Topic  
= 2

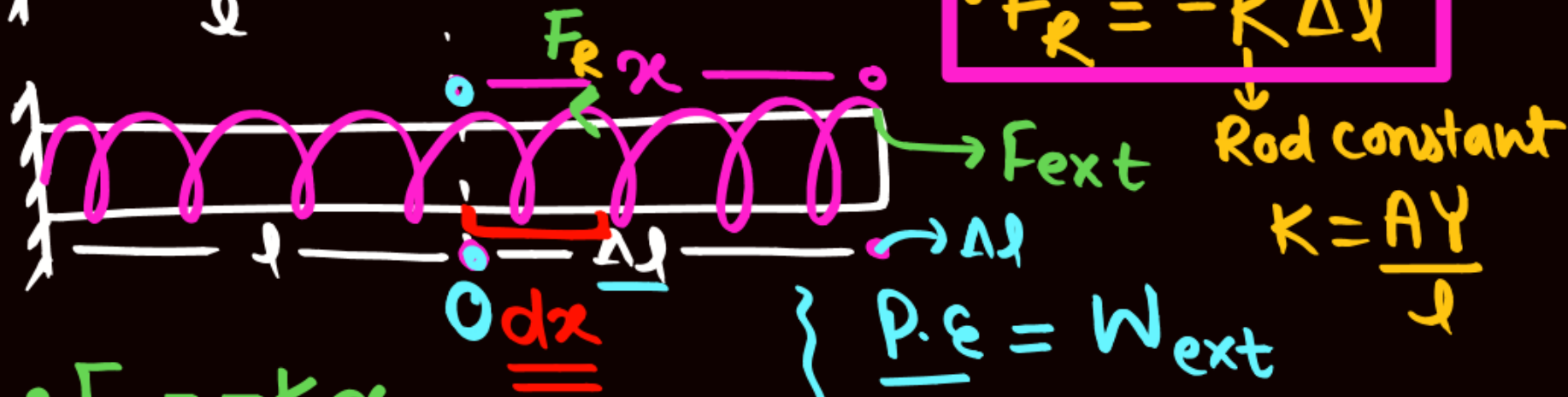
# Elastic Potential Energy in Rod (Same / Same $\Rightarrow$ P.E store $\Delta l$ in spring)

# The p.e store when you stretched the Rod



$$\bullet F_s = -kx$$

$$\bullet F_R = -k\Delta l$$



$$\bullet F_R = -kx$$

$$\# F_{ext} \geq F_R$$

$$F_{ext} = kx$$

$$\underline{P.E} = W_{ext}$$

$$W = \underline{F}x$$

$\Downarrow$  variable

$$\underline{dw} = Fdx$$

$$\int dw_{ext} = \int F_{ext} dx$$

$$W_{ext} = \int_0^{\Delta l} kx dx$$

$$W_{ext} = k \int_0^{\Delta l} x dx$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$W_{ext} = k \left[ \frac{x^{1+1}}{1+1} \right]_0^{\Delta l}$$



$$W_{ext} = K \left[ \frac{x^2}{2} \right]_0^{\Delta l}$$

$$W_{ext} = K \left[ \frac{\Delta l^2}{2} - \frac{0^2}{2} \right]$$

$$\underline{W_{ext}} = \underline{K} \frac{\Delta l^2}{2} \quad K = \frac{\gamma A}{l}$$

$$\underline{W_{ext}} = \frac{\gamma A}{l} \frac{\Delta l^2}{2}$$

$$U = \frac{1}{2} \gamma \underline{A} \frac{\Delta l^2}{l} \quad \underline{A \times l = V}$$

divide and multiple (l) on  
RHS

$$U = \frac{1}{2} \gamma \underline{A} \frac{\Delta l^2}{l} \times \underline{\frac{l}{l}}$$

$$U = \frac{1}{2} \gamma \underline{V} \frac{\Delta l^2}{l^2} \quad \frac{\Delta l}{l} = \text{Strain}$$

$$\frac{U}{V} = \frac{1}{2} \gamma \left( \frac{\Delta l}{l} \right)^2 \quad \frac{\Delta l^2}{l^2} = \text{Strain}^2$$

$$\frac{U}{V} = \frac{1}{2} \gamma \text{Strain}^2$$

$$\frac{U}{V} = +\frac{1}{2} \times \gamma \times \text{Strain}^2 \quad \text{--- ①}$$

?

$$\gamma = \frac{\text{Stress}}{\text{Strain}}$$

$$\gamma = \frac{\text{Stress}}{\text{Strain}}$$

$$\text{Strain} = \frac{\text{Stress}}{\gamma}$$

$$\frac{U}{V} = \frac{1}{2} \gamma \text{Strain}^2$$

$$\frac{U}{V} = \frac{1}{2} \frac{\text{Stress}}{\text{Strain}} \text{Strain}^2$$

$$\frac{U}{V} = \frac{1}{2} \text{Stress} \times \text{Strain} \quad \text{--- ②}$$

$$\frac{U}{V} = \frac{1}{2} \times \gamma \times \text{Strain}^2$$

$$\frac{U}{V} = \frac{1}{2} \times \gamma \times \left( \frac{\text{Stress}}{\gamma} \right)^2$$

$$\frac{U}{V} = \frac{1}{2} \times \cancel{\gamma} \times \frac{\text{Stress}^2}{\cancel{\gamma}}$$

$$\frac{U}{V} = \frac{1}{2} \frac{\text{Stress}^2}{\gamma} \quad \text{--- ③}$$



## Poisson Ratio (2 mark)

# It is the negative Ratio  $\frac{\text{Lateral Strain}}{\text{or Shear Strain or tangential Strain}}$  to the longitudinal Strain

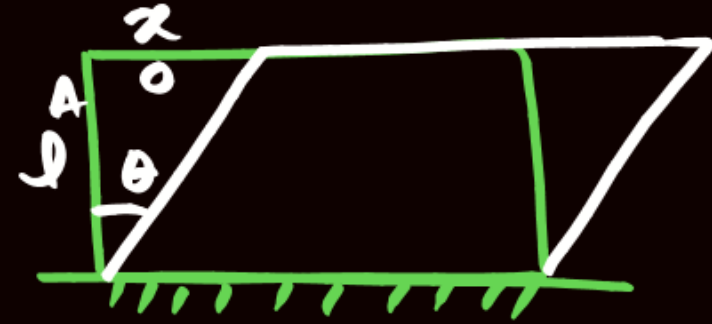
$$\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \times \frac{d}{c}$$

$$PR = - \frac{\text{Shear Strain}}{\text{Longitudinal Strain}}$$

$$PR = - \frac{\tan \theta}{\frac{\Delta l}{l}}$$

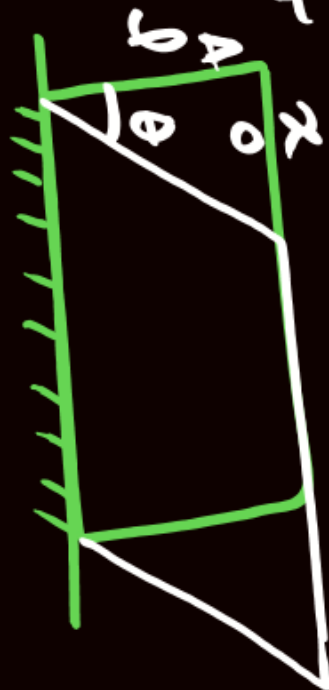
$$PR = - \frac{\frac{x}{l}}{\frac{\Delta l}{l}}$$

$$PR = - \frac{x}{l} \times \frac{l}{\Delta l} = - \frac{x}{\Delta l} \quad \checkmark$$



$$\tan \theta = \frac{x}{l}$$

$$\tan \theta = \frac{x}{l}$$



# HOMEWORK



Notes → community

Derivation  $\Rightarrow$  Repeat.

“The secret to getting ahead is  
getting started.”





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**Thank**  
*You*

